Matrix inequalities and norms

Lieb's permanental dominance conjecture

Difficulty: extreme **Importance:** interesting to the community

Status: explicit conjecture; no general resolution located

Let $n \geq 1$, let $A = (a_{ij}) \in \mathbb{C}^{n\times n}$ be Hermitian positive semidefinite, let G be any subgroup of the symmetric group S_n, and let χ be the character of any nonzero finite-dimensional complex representation of G. Thus $\chi(e) > 0$, where e is the identity. Define

$$f_\chi(A) = \frac{1}{\chi(e)} \sum_{\sigma\in G} \chi(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)}, \qquad \operatorname{per} A = \sum_{\sigma\in S_n} \prod_{i=1}^{n} a_{i,\sigma(i)}.$$

Does the inequality

$$f_\chi(A) \leq \operatorname{per} A$$

hold for every such n, A, G, χ? The character expression is real for Hermitian A, so the comparison is an ordinary real inequality.

Relevance

Generalized matrix functions include the determinant and immanants. The conjecture seeks a common upper bound across these structured matrix polynomials on the positive-semidefinite cone. It is part of matrix computation and matrix inequality theory, with links to symmetry reductions of tensor powers.

References

- E. H. Lieb, *Proofs of some Conjectures on Permanents*, Journal of Mathematics and Mechanics 16 (1966), 127–134, Conjecture α on p.127 (publisher's first page).
- I. M. Wanless, *Lieb's permanental dominance conjecture* (2022), Conjecture 1, §2 implication diagram, and §3 partial results (primary manuscript).
- A. Rico, D. Grinko, R. Krebs, and L. H. Zaw, *Entanglement Structure and Matrix Inequalities from Isotypic Measurements*, Physical Review Letters 137 (2026), 100203; abstract describes immanant inequalities for orders three and four (journal).

Status check — 2026-09-08

Searches for the exact conjecture name and "proof", "counterexample", and 2025–2026 found no general resolution. The September 2026 PRL abstract describes fixed-order progress. The August 2026 preprint *GPT, the Counterexample Machine*, §3.1, refutes a stronger conjecture for arbitrary real-stable polynomials; its example is not presented as a PSD-matrix counterexample to Lieb's conjecture (primary manuscript). Likewise, the older disproof of permanent-on-top does not resolve the displayed statement.